Department of Engineering Mathematics

Constraining The composition of Earth's core: Using ensemble models to predict metal-silicate partitioning from experimental data

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A Master's thesis submitted to the University of Bristol
in accordance with the requirements of the degree of Master of Science
in the Faculty of Engineering

Wednesday 25th Feb, 2026

Abstract

The composition of Earth's core remains a fundamental but unresolved question in Earth and planetary sciences. Seismological evidence indicates that while the core is primarily an iron–nickel alloy, it is several per cent less dense than pure Fe–Ni. This discrepancy implies the presence of lighter elements, with candidates including silicon, oxygen, hydrogen, carbon, and sulphur. Establishing the relative proportions of these elements is of central importance: it provides constraints on the processes of planetary accretion and differentiation, governs the chemical and thermal exchanges between mantle and core, and influences the long-term stability of the Earth's magnetic field.

This dissertation investigates core composition by integrating geophysical observations, high-pressure geochemical experiments, and cosmochemical models within a data science framework. Laboratory studies using diamond anvil cells, multi-anvil presses, and piston-cylinder apparatuses provide essential partitioning data for siderophile and lithophile elements under extreme pressure–temperature conditions. These experimental results are complemented by computational approaches, including first-principles calculations, and by comparisons with the compositions of chondritic meteorites, which act as proxies for Earth's primordial building blocks.

By synthesising these strands of evidence, the project develops a model of the Earth's core that reconciles seismic constraints with geochemical budgets. The findings suggest that no single light element can account for the observed density deficit. Instead, the most consistent explanation is a mixture dominated by silicon, oxygen, and hydrogen. This outcome not only refines current estimates of core composition but also highlights the methodological value of combining experimental, computational, and observational datasets through data science techniques. In doing so, the study contributes to resolving one of the most enduring questions about the Earth's interior, while offering a transferable framework for investigating the inaccessible cores of other planetary bodies.

Keywords: Earth's core; planetary differentiation; siderophile and lithophile elements; light elements; high-pressure experiments; seismic constraints; data science in geoscience

Dedication and acknowledgements

I would like to give my first and foremost thanks to my thesis supervisor,Dr. Simon Lock, in the School of Earth Sciences at the University of Bristol. His guidance, encouragement, and expertise have been invaluable throughout the course of this project. He was always generous with his time, providing clarity when I encountered challenges in the research and offering insightful suggestions that pushed me to think more critically and ambitiously. His ability to connect geochemical concepts with broader geophysical implications shaped the direction of this work and helped me to develop a deeper understanding of the field. I am also deeply grateful for his support in identifying relevant literature, sharing resources, and offering constructive feedback on my writing and analysis. His mentorship has not only enhanced the quality of this dissertation but has also had a lasting impact on my academic development.

Beyond my supervisor, I would also like to thank James Pope, my additional supervisor, for his support

Lastly, I recognize the steadfast encouragement from my family, friends, and partner. Their understanding, comfort, and unwavering confidence in my abilities gave me the drive necessary to see this work through to completion. Without their persistent support throughout the most challenging phases of this research, this thesis could not have been realized.

Author's declaration

This dissertation is submitted to the University of Bristol in accordance with the requirements of the degree of Master of Science in the Faculty of Engineering. It has not been submitted for another degree or diploma at any examining body. Except where specifically acknowledged, this work represents the original research of the Author.

SIGNED: Radha Prashant Kadam DATE: 25th Feb, 2026.

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Chapter 1

Introduction

Earth's core composition represents one of the most fundamental yet poorly constrained aspects of our planet's structure and evolution. Comprising approximately 32% of Earth's mass and extending from 2900 km depth to the planetary centre, the core controls magnetic field generation, heat flow patterns, and long-term planetary evolution through its chemical and physical properties. Despite decades of experimental and theoretical research, significant uncertainties remain regarding the identity and concentrations of light elements that account for the core's observed density deficit relative to pure iron-nickel alloys.

Traditional approaches to determine the core composition rely on experimental metal-silicate equilibrium studies conducted at high-pressure, high-temperature conditions intended to simulate planetary accretion processes. However, they mostly focus on individual elements or limited element suites, providing segmented constraints that cannot be directly integrated into planetary formation models[9]. Essentially, these methods cannot accurately predict the complete core compositions because they are unable to capture the complex, multi-element interactions that happen during planetary differentiation. The resulting uncertainty in the composition propagates through modelling of the Earth's magnetic field, affecting the predictions of core thermal evolution, magnetic field behaviour, and planetary differentiation timescales.

Recent advanced machine learning models offer unique opportunities to integrate different experimental datasets, capturing complex multi-element interactions that traditional analytical approaches have not adequately addressed. This research aims to develop computational methods that can predict near-complete compositions from experimental conditions, whilst maintaining thermodynamic consistency ,and geochemical validity. This also addresses the need for comprehensive approaches moving beyond the limitations of fragmented, single-element studies toward integrated, multi-element compositional predictions.

1.1 Research Challenges and Scientific Gaps

The prediction of Earth's core composition from experimental conditions presents multiple interconnected challenges that have limited progress in computational geochemistry. Experimental datasets remain relatively small due to the technical complexity and cost of high-pressure metal-silicate equilibration studies, typically

providing 150-300 samples across limited pressure-temperature ranges. This data dearth constrains the application of sophisticated machine learning algorithms that require large training datasets for robust performance. Element concentration ranges span six orders of magnitude from major constituents (Fe: 70-95%) to ultra-trace species (Nb: <0.01%), creating computational challenges for algorithms designed to handle homogeneous data distributions. Traditional statistical approaches struggle with this heterogeneity, while machine learning methods require careful algorithm selection and optimisation to achieve meaningful results across all concentration levels. The multi-factorial nature of element partitioning during core formation involves complex interactions between pressure, temperature, oxygen fugacity, sulphur fugacity, carbon fugacity, and bulk composition parameters that influence individual element behaviour through thermodynamic activity relationships. Previous computational studies have typically simplified this complexity by focusing on pressure-temperature inputs alone, limiting their applicability to experimental datasets where fugacity conditions are explicitly controlled. Previous studies have predicted only a few core elements, leaving 20-40% of the core's composition unknown. This incomplete picture prevents more accurate calculation of thermodynamic properties and their comparison with seismological data on core density and velocity structure[21]

1.2 Investigative Focus and Research Objectives

This research develops a machine learning framework to address three fundamental challenges constraining computational geochemistry. First, **incomplete elemental coverage** occurs because existing approaches predict only 60-80% of Earth's core mass, leaving critical gaps in light elements (H, C, P) and ultra-trace refractory species that prevent complete mass balance calculations essential for thermodynamic modelling. Second, **algorithm inadequacy** stems from extreme dataset heterogeneity where element concentrations span six orders of magnitude from major constituents (Fe: 70-95 wt%) to ultra-trace species (Nb: <0.01 wt%), creating computational challenges that traditional statistical methods cannot handle due to assumptions of homogeneous data distributions. Third, **limited experimental integration** prevents incorporation of fugacity parameters (fO2, fS2, fC) that control redox conditions in high-pressure experiments, limiting applicability to controlled studies providing the most reliable partitioning data. The framework establishes three interconnected objectives addressing these limitations.

Complete core composition prediction will be achieved for fourteen elements (Fe, Ni, O, S, Si, C, H, P, Cr, Co, W, Mo, V, Nb) representing 100% of core mass, with perfect closure constraints. This capability has remained elusive despite decades of computational efforts. Element-specific modelling strategies will accommodate distinct partitioning behaviours, recognising that light volatile elements (H, C, P) follow fundamentally different thermodynamic pathways than siderophile constituents (Fe, Ni, Co) or ultra-trace, refractory elements (W, Mo, V, Nb), which require specialised algorithmic treatments. Fugacity-aware feature engineering will enable direct application to modern experimental studies where oxygen, sulphur, and carbon fugacities are precisely controlled. Current data limitations include insufficient experimental coverage of light element behaviour under core formation conditions (25-70 GPa, 2500-4000 K) and limited data for refractory elements due to analytical challenges in measuring ultra-trace concentrations, making W, Mo, V, and Nb the least understood core components. This framework provides both practical tools for experimental validation and methodological innovations applicable to planetary differentiation processes.

CHAPTER 2

Background and Literature Review

2.1 Earth's Core Formation and Compositional Constraints

Earth's core formation represents one of the most significant differentiation events in planetary history, fundamentally shaping our planet's subsequent thermal, magnetic, and geochemical evolution. The process occurred during the first 30-100 million years following planetary accretion, when gravitational energy release and radioactive decay provided sufficient thermal energy to initiate large-scale metal-silicate segregation. Understanding this process requires quantitative knowledge of element partitioning behaviour under the extreme pressure-temperature conditions that characterised early Earth's interior.

Ringwood (1959)[20] proposed that Earth's core formed through carbon reduction of primitive oxidised silicate material during planetary melting, producing a metallic iron-nickel-silicon phase that descended to form the core. He estimated the core contains around 20% silicon, which is better considered for its observed density than pure nickel-iron alone. Jones and Drake (1986)[13] stated that all viable models of core formation require the presence of a light element in the core to account for its observed density deficit relative to pure iron-nickel, with silicon, sulphur, and carbon identified as the most reasonable candidates.

The modern understanding of core composition emerged from the integration of experimental metal-silicate partitioning data with cosmochemical constraints and seismological observations. McDonough and Sun [16]established that Earth's core must contain 5-15% light elements by mass to explain the observed density deficit relative to pure iron-nickel alloys, while seismological studies by Dziewonski and Anderson [7] provided precise constraints on core density and seismic velocity structure that compositional models must satisfy.

2.2 Experimental Approaches to Core Composition Determination

High-pressure experimental geochemistry provides the primary methodology for quantifying element partitioning behaviour during core formation. The experimental approach involves equilibrating metal and silicate phases at controlled pressure-temperature-oxygen fugacity conditions, followed by compositional analysis to determine metal-silicate distribution coefficients. This methodology was advanced by Walker et al. (1993)[23], who performed

high-pressure metal-silicate partitioning experiments and demonstrated that mantle abundances of moderately siderophile elements such as Ni, Co, W, and Mo are consistent with core-mantle equilibration at elevated pressures and temperatures.

The critical role of pressure was established by Li and Agee (1996)[15], who showed using multi-anvil press experiments that Ni and Co partition coefficients converge at high pressure, providing the primary experimental basis for the deep magma ocean model of core formation. Their work demonstrated that core-mantle equilibration at pressures of 25–30 GPa could reproduce observed mantle siderophile abundances. Fischer et al. (2015)[8] extended these investigations using both multi-anvil press and diamond anvil cell experiments to 100 GPa and 5700 K, parameterising the effects of pressure, temperature, and metallic melt composition on the partitioning of Ni, Co, V, Cr, Si, and O ScienceDirect, demonstrating that temperature effects become increasingly important at extreme conditions.

2.2.1 Fugacity Control and Redox Systematics

The recognition that oxygen fugacity exerts fundamental control on element partitioning during metal-silicate equilibration represents a critical advancement in experimental core formation studies. O'Neill (1991)[17] proposed that metal-silicate distribution coefficients for multivalent elements depend strongly on oxygen fugacity, establishing the theoretical framework for fugacity-controlled experiments that enable systematic investigation of redox effects on core composition.

Wade and Wood (2005)[22] quantified oxygen fugacity effects across ranges relevant to planetary accretion, demonstrating that core formation occurred under progressively increasing fO2 conditions as the Earth grew, with silicon and oxygen contents of the core strongly sensitive to redox state. Their work established oxygen fugacity as a primary control on core composition during planetary formation.

Extended fugacity studies demonstrated that sulphur and carbon fugacities also influence light element incorporation during core formation. Dasgupta and Walker (2008)[6] quantified carbon solubility in core melts across a range of pressure and temperature conditions, while Dasgupta et al. (2013)[5] demonstrated that carbon partitioning between metallic and silicate melt depends strongly on oxygen fugacity, temperature, and metallic melt composition. These studies collectively established that multiple fugacity variables must be considered for comprehensive core composition modelling.

2.3 Computational Approaches and Machine Learning Applications

Computational methods in geochemistry began with statistical analysis of experimental partitioning data. Jones and Malvin (1990)[14] developed thermodynamic activity corrections for trace metal segregation in Fe-Ni systems, demonstrating that systematic relationships could be extracted from partitioning experiments. Righter et al. (1997) [19]built on this by creating multi-variable regression models that simultaneously parameterise the effects of pressure, temperature, and oxygen fugacity on siderophile element partitioning, an approach that remains standard in core formation modelling today.

More recently, machine learning has begun to transform geochemical data analysis. He et al. (2022)[10] showed that machine learning can reveal hidden patterns in complex geochemical datasets, with applications spanning rock classification, environmental monitoring, and planetary science. Breiman's (2001)[3] Random Forest algorithm has proven particularly effective for high-dimensional problems of this kind, offering strong predictive accuracy across varied scientific domains.

Despite these advances, no published study has applied machine learning specifically to predicting Earth's core composition from metal-silicate partitioning data. Existing computational approaches still rely on thermodynamic equilibrium calculations anchored to experimental data, as established by Wade and Wood (2005)[22] and Righter et al. (1997)[19]. This gap, driven by limited dataset sizes, extreme compositional ranges, and complex element interactions, provides the primary motivation for the present work.

2.4 Theoretical Framework for Element Partitioning

The thermodynamic theory of element partitioning during metal-silicate equilibration states the physical foundation for validating and constructing machine learning models. The equilibrium distribution of an element between metallic and silicate phases is expressed as:

$$K_D = \frac{X_{metal}}{X_{silicate}} \times \frac{\gamma_{silicate}}{\gamma_{metal}} [1]. \tag{2.1}$$

where K_D is the distribution coefficient, X denotes mole fractions, and γ represents activity coefficients that account for non-ideal solution behavior.

Experimental work by Holzheid *et al.* (1997) [12] demonstrated that distribution coefficients exhibit systematic dependencies on pressure (P) and temperature (T). These relationships follow the form:

$$\log K_D = A + \frac{B}{T} + C \times \frac{P}{T} [12] \tag{2.2}$$

where A, B, and C are element-specific constants determined empirically. This framework directly guides machine learning feature engineering. Pressure, temperature, and inverse temperature terms are the physically meaningful variables expected to capture key aspects of partitioning behaviour.

Oxygen fugacity effects were quantified by Wade and Wood (2005)[22], who demonstrated that fO2exerts primary control on the metal preference of multivalent elements through systematic modifications to activity coefficients. O'Neill and Mavrogenes (2002) [18]extended thermodynamic constraints to sulphur fugacity, establishing solubility relationships in silicate melts under variable redox conditions. Element interaction effects in metallic liquids are described through the Wagner interaction parameter formalism as refined by Bale and Pelton (1990)[2], which quantifies how dissolved elements mutually influence each other's activity coefficients.

Together, these relationships define the physically constrained feature space within which machine learning predictions must remain geochemically valid.

2.5 Current State of Knowledge and Research Gaps

Despite substantial progress in experimental and theoretical understanding of core formation processes, significant gaps remain in our ability to predict complete core compositions from experimental conditions. Current experimental

studies typically focus on individual elements or limited element suites, preventing a comprehensive assessment of multi-element interactions that influence realistic core compositions. The resulting fragmented compositional data cannot be directly integrated into planetary formation models requiring a complete elemental mass balance.

Existing computational approaches suffer from several limitations that constrain their applicability to comprehensive core composition modelling. Most studies utilise only pressure and temperature inputs, ignoring fugacity controls that experimental studies demonstrate as essential for accurate partitioning prediction. The absence of systematic algorithm optimisation results in suboptimal performance for different element types, while the lack of ensemble approaches prevents the exploitation of complementary algorithmic strengths.

The limitation to partial composition prediction represents perhaps the most significant constraint on current computational geochemistry applications. Studies achieving only partial composition coverage cannot provide the complete elemental accounting required for thermodynamic property calculation or direct comparison with seismological observations. This limitation prevents robust validation of planetary formation models and constrains the applicability of computational results to broader Earth science applications.

CHAPTER 3

Data Analysis and Preprocessing

3.1 Dataset Description and Characteristics

The study draws on a wide range of Earth's core composition measurements, gathered from several high-pressure experimental experiments, consisting of 196 experimental samples with 208 variables, including pressure-temperature conditions, elemental compositions, and some related uncertainties. Experimental environments cover pressures that vary from 1.0 to 100.0 GPa and temperatures from 1,773 to 5,700 K, effectively specifying the pressure-temperature spectrum essential to Earth's core formation, from shallow magma ocean conditions to core-mantle boundary extremes. Each experimental entry provides extensive partitioning behaviour data for major elements (Fe, Si, O, S) and trace elements (Al, Mg, Ca, among others), with measurements reflecting mole fractions in metallic and silicate phases that offer quantitative constraints on element distribution under numerous pressure-temperature conditions during planetary differentiation.

3.2 Data Preprocessing and Quality Control

3.2.1 Data Cleaning Methodology

To ensure the reliability of the analysis, the raw data had to undergo preprocessing. The procedure was designed to address frequent problems in assembled experimental datasets while preserving their scientific value. The greatest challenge was handling the non-numeric entries that occurred when compiling data and performing calculations in spreadsheets. In 847 instances, there are some Excel error codes like "VALUE!", "DIV/0!", "N/A", "REF!", etc., because of some failed math operations or data references were missing. These error codes were systematically identified and replaced with NaN (Not a Number) values to prevent computational errors during analysis while maintaining the statistical integrity of the remaining data.

A crucial aspect involved treating "below detection limit" (`bdl` or `BDL`) values, which are common in trace element analysis where element concentrations fall below analytical detection thresholds of measurement techniques. Rather than treating these as missing data and losing valuable information, a total of 78 `bdl` entries

were replaced with 1×10^{-8} following established geochemical practices, preserving the scientific information that elements were present but undetectable while allowing these entries to participate in statistical calculations.

Finally, structural optimisation removed many empty columns with names like "Unnamed" followed by numbers or those entirely composed of underscores. These columns were artefacts from the data compilation process and didn't add any analytical value. This reduced the dataset from 208 to 137 meaningful variables, which significantly accelerated the calculations while retaining all scientifically relevant information, including experimental conditions, elemental compositions, measurement uncertainties, and study identification data.

3.2.2 Data Quality Assessment

Statistical evaluation confirms exceptional data completeness at 98.7% valid numerical values after preprocessing. Inter-study consistency validation shows remarkable reliability across compiled datasets, with overlapping experimental conditions from different groups agreeing within reported uncertainties. This validates both experimental techniques and compilation methodology.

3.3 Exploratory Data Analysis

3.3.1 Pressure-Temperature Experimental Coverage

Figure 3.1demonstrates comprehensive coverage of pressure-temperature space related to Earth's core formation. Data points form coherent trends from low conditions (~1 GPa, 1800 K) to extreme core-mantle boundary conditions (~100 GPa, 5500 K). The colour gradient illustrates a systematic increase in pressure with temperature correlation, following realistic geothermal profiles during magma ocean evolution.

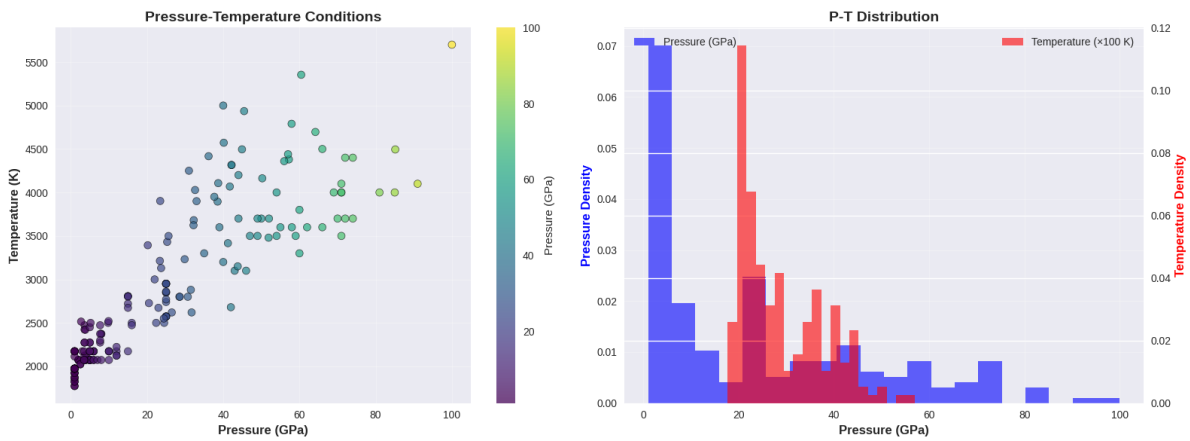

FIGURE 3.1. a)Pressure-Temperature Conditions of Analysed Samples b) P-T Distribution (Frequency histograms for pressure and temperature measurements).

Experimental coverage shows dense sampling at intermediate pressures (20-60 GPa), corresponding to critical depths where major core differentiation occurred. Histogram analysis reveals that pressure distribution peaks at lower pressures, with a decreasing frequency at extremes, reflecting technical challenges and a scientific focus on

shallow processes. The temperature distribution exhibits a broader, more uniform coverage across the experimental ranges.

3.3.2 Correlation Structure and Elemental Interactions

Figure 3.2 illustrates the complex interrelationships that govern the elemental partitioning. There is a good positive correlation (r = 0.88) between the pressure and temperature, which implies that realistic geothermal gradients have been explored systematically, not by chance.

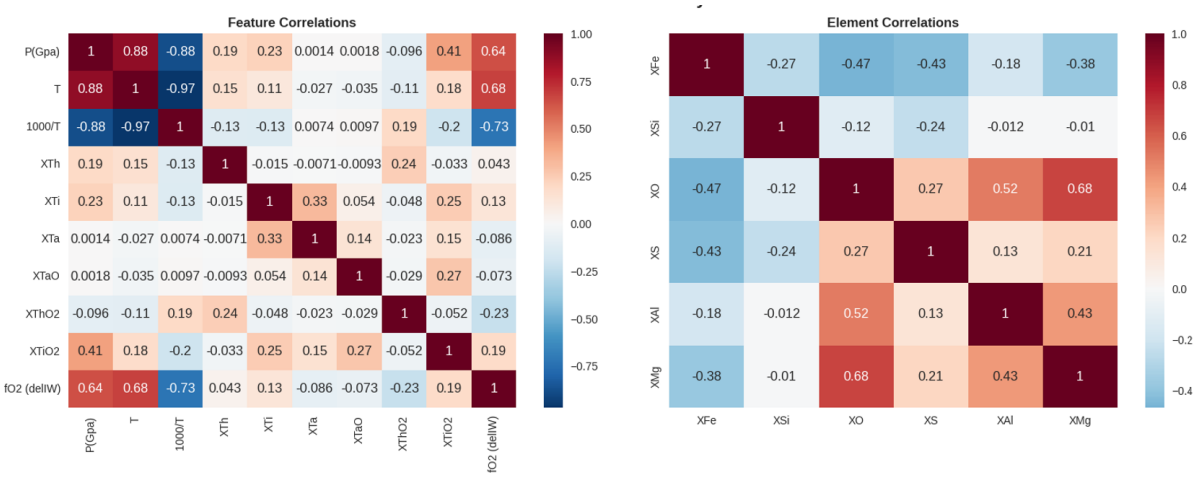

FIGURE 3.2. a)Feature Correlation Matrix: P-T Conditions and Geochemical Variables b) Major Element Correlation Matrix: Oxide Chemistry Patterns

Oxygen fugacity (fO_2) shows moderate positive correlations with pressure (r = 0.64) and temperature (r = 0.68), suggesting more oxidising conditions at higher pressure-temperature conditions. Element correlations show iron (X_{Fe}) negatively correlating with light elements, particularly silicon (r = -0.27) and magnesium (r = -0.38), indicating competitive partitioning behaviour. Oxygen and magnesium display a strong positive correlation (r = 0.68), suggesting coupled partitioning mechanisms.

3.3.3 Comprehensive Correlation Analysis

Correlation analysis quantifies linear relationships using correlation coefficients (r) ranging from -1 to $+1$. Most individual correlations remain weak ($r < 0.5$), indicating that simple linear relationships cannot capture the complexity of core formation. This justifies advanced feature engineering for machine learning models. The correlation matrix shows the expected thermodynamic patterns but not the expected patterns of complex process interactions. A high correlation of pressure and temperature justifies systematic experimental design obeying the realist conditions, rather than random combinations.

Data Completeness Limitations: The matrix shows significant analytical constraints through systematic missing values, particularly the complete absence of correlations for elements like sodium (Na) and several other trace components, represented by blank white spaces in Figure 3.3. These gaps indicate that these elements remained consistently below analytical detection limits across all experimental conditions, reflecting either a

genuine absence of incorporation into metallic phases under the studied pressure-temperature range or fundamental limitations of the analytical techniques employed (electron microprobe, LA-ICP-MS). This pattern is scientifically meaningful as it demonstrates the detection threshold boundaries for different elements under extreme experimental conditions, with highly lithophile elements like sodium showing no measurable partitioning into metallic phases during metal-silicate equilibration experiments.

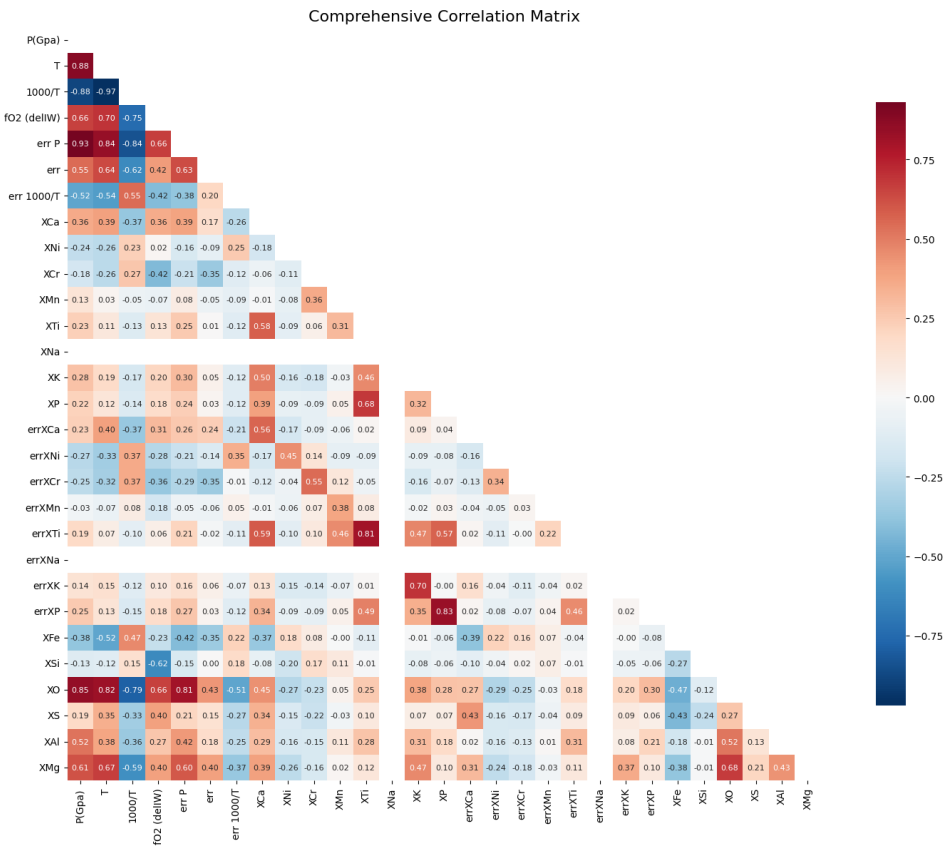

FIGURE 3.3. Comprehensive Correlation Matrix of Experimental Variables and Elemental

The correlation matrix reveals expected thermodynamic relationships and unexpected patterns indicating complex process interactions. Strong pressure-temperature correlation confirms systematic experimental design following realistic conditions rather than arbitrary combinations.

3.4 Principal Component Analysis and Dimensionality Reduction

3.4.1 Variance Structure Analysis

Principal Component Analysis (PCA) applied to engineered features identifies underlying variance patterns in geochemical measurements, reducing dimensionality while preserving essential information and determining important factors controlling compositional variations during core formation.

The hierarchical structure reveals the relative importance of different processes. The first principal component dominates the variance structure, explaining 30.0% of total variation, representing primary control mechanisms.

The second component contributes 15.6%, followed by the third at 9.4%, creating a characteristic exponential decay typical of structured geochemical datasets.

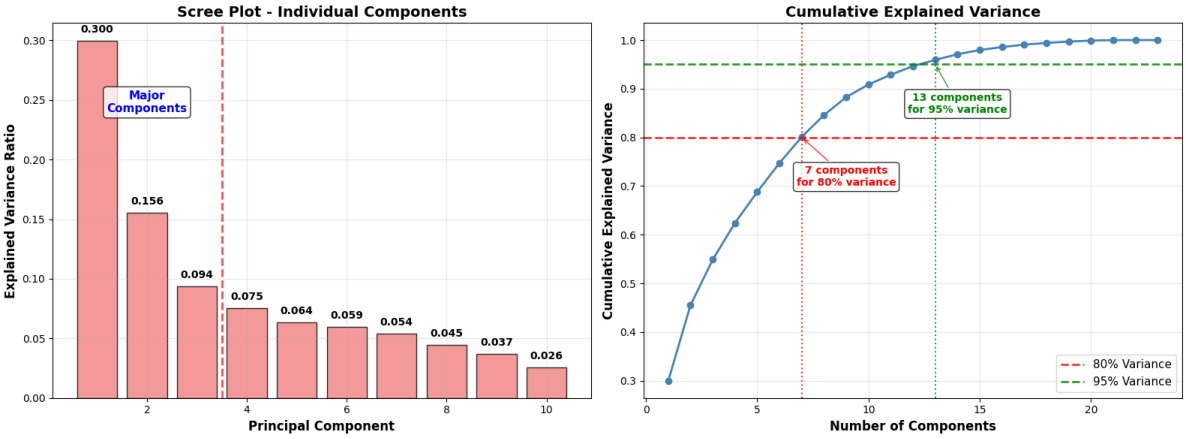

Figure 3.4: a) Individual Principal Component Contributions b) Cumulative Variance Explained with 80% and 95% Thresholds

This hierarchy reveals clear process importance: the dominant first component represents primary pressure-temperature controls, the second suggests oxygen fugacity and redox controls, while the third captures interaction effects between pressure, temperature, and composition variables.

3.4.2 Dimensionality Reduction Strategy

Cumulative variance analysis demonstrates efficient dimensionality reduction while preserving essential information. Seven components capture 80% of total variance, achieving a significant reduction from 24 engineered features. Thirteen components capture 95% variance, maintaining nearly complete information content.

3.4.3 Spatial Distribution Patterns

PCA score plots reveal the organisation of experimental samples in a reduced-dimensional space, providing insights into the underlying geochemical processes controlling core formation.

Primary Process Control (PC1 vs PC2): Higher pressure-temperature experiments cluster in positive PC1 regions, while lower P-T conditions occupy negative PC1 space. Colour-coded temperature gradients (1800K purple to 5500K yellow) show samples arranged along temperature-controlled trajectories, validating coherent P-T space coverage.

Secondary Process Identification (PC2 vs PC3): Additional structure through iron content variations shows samples with higher iron content cluster in distinct regions independent of primary P-T controls. This separation indicates iron partitioning follows systematic rules operating orthogonally to pressure-temperature processes.

3.4.4 Modelling Implications

A clear hierarchical structure provides quantitative guidance for dimensionality reduction in predictive modelling. A steep decline in component variance after the third component, combined with a cumulative variance plateau around eight components, establishes natural breakpoints for model complexity decisions.

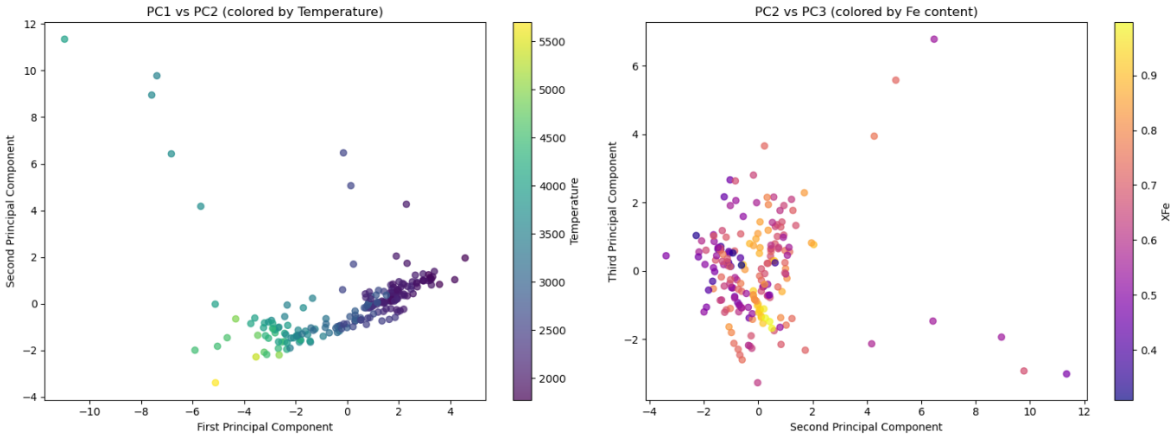

FIGURE 3.5. a) PC1 vs PC2 - Primary Geochemical Variation (coloured by Temperature) b) PC2 vs PC3 - Secondary Geochemical Variation (coloured by Fe Content)

Successful dimensionality reduction from 24 engineered features to 5-8 meaningful components provides substantial computational advantages while preserving essential information content. This efficiency indicates complex core formation processes are controlled by relatively few systematic factors, with remaining variance representing minor secondary effects or experimental uncertainties.

3.5 Dataset Readiness for Machine Learning Applications

The evaluated and processed data demonstrate exceptional accuracy in complicated computational modelling. Systematic preprocessing effectively addressed data quality issues while preserving scientific integrity. In organised, well-defined data with a thorough representation of the parameter space. The effective combination of multiple independent research groups, along with inter-study validation. Consistency ensures that the dataset accurately represents the current knowledge of element partitioning in severe circumstances. The hierarchical variance structure shown by PCA analysis shows that the dataset contains predictable patterns suitable for machine learning applications and is sufficiently substantial. Complexity requires sophisticated modelling. The mathematical connections between components and original variables are clear, facilitating an interpretable model. Structures that maintain scientific significance and exhibit analytical efficiency, which provide the foundation of. Effective forecasting techniques enhance understanding of the differentiation and evolution of planetary cores.

CHAPTER 4

Feature Engineering and Advanced Variable Construction

Feature engineering is a crucial preprocessing stage in ML algorithms, which helps convert experimental data into a form that can be used to model the physics and chemistry of the metal-silicate equilibrium. Statistically consistent variables can reflect a complicated non-linear association between elements present in planetary core formation. This may not be restricted to minor transformations in data to include geochemical expertise of high-pressure processes. The need for feature engineering in core formation investigations comes from the realisation that elemental partitioning behaviour is severely complicated. The dependence on experimental conditions means that linear relationships among individual variables cannot be precisely modelled. The thermodynamic principles state that partition coefficients are product- and ratio-dependent.

4.1 Feature Engineering Methodology and Physical Basis

This approach generates polynomial transformations, interaction terms, and mathematical relationships which is proportional to thermodynamic controls on element partitioning. Pressure-temperature interaction terms ($P \times T$) capture combined effects on phase equilibria, as thermodynamic theory shows that many equilibrium constants depend on $P \times T$ products rather than individual values. Polynomial terms (P^2, T^2) are associated with non-linear threshold effects observed in high-pressure systems. Simultaneously, logarithmic transformations ($\log P$, $\log T$) linearise exponential relationships from Arrhenius-type equations governing reaction kinetics and equilibrium constants. The inverse temperature term ($1000/T$) aligns well with standard Arrhenius plot representations widely used in geochemistry.

4.2 Implementation of Advanced Variable Construction

The feature engineering systematically expanded four basic parameters (pressure, temperature, oxygen fugacity, inverse temperature) to 24 engineered variables through mathematical transformations built to capture non-linear relationships whilst retaining computational efficiency. Primary interaction terms include pressure-temperature products that represent fundamental coupling controlling phase stability, and oxygen fugacity interactions with both pressure and temperature, normalising combined redox and physical condition effects on element speciation.

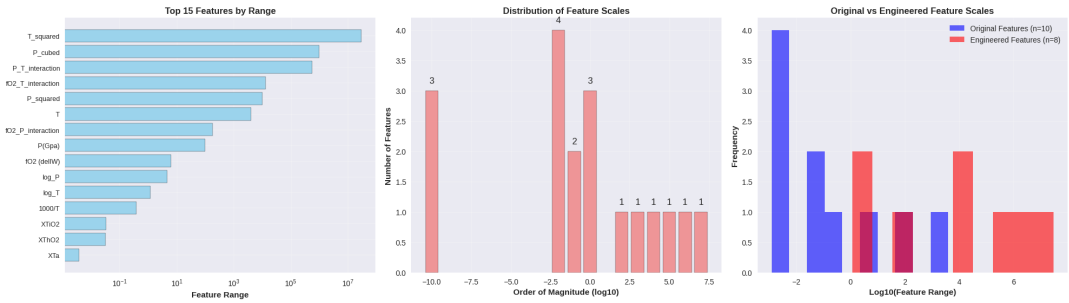

FIGURE 4.1. (a)Feature Range Analysis of Top 15 Variables by Scale Magnitude (b) Feature Scale Distribution: Frequency of Variable Range Values (c) Original vs Standardised upperbound Effectiveness Feature Ranges

Polynomial transformations incorporate second-order terms (P^2, T^2), which account for threshold effects and non-linear responses. Whereas, third-order terms (P^3) also facilitate capturing complex saturation behaviours under extreme conditions approaching core-mantle boundary pressures. Mathematical function transformations include square root terms ($\sqrt{P}$, $\sqrt{T}$), which provide intermediate non-linearity, and logarithmic terms ($\log P$, $\log T$), that linearise exponential dependencies, creating a mathematical basis set representing functional relationships. Ratio variables, such as pressure-temperature ratios (P/T, T/P), represent normalised conditions controlling equilibrium positions independently of absolute values, particularly relevant for understanding how experimental conditions correspond to natural core formation P-T evolutionary paths.

4.3 Feature Scale Analysis and Normalisation Requirements

4.3.1 Scale Disparity Assessment

The transformation process inculcates observable variations in feature scale, requires systematic normalisation for an effective ML approach Original pressure measurements between 1 and 100 GPa become pressure-squared terms ranging from 1 to 10,000, which extend to pressure-cubed features of 1,000,000, creating a six orders of magnitude difference between basic measurements and their polynomial transformations. Similarly, temperature values ranging from 1,800 to 5,500 K generate temperature-squared terms reaching 30,000,000, stating the critical importance of systematic scaling.

Figure 4.1 indicates these scale disparities across the complete feature set. The left panel identifies the most problematic features ranked by numerical range, revealing that T_{squared} and P_{cubed} display the largest ranges exceeding 10^6, while basic experimental measurements like X_{ia} and X_{TiO_2} occupy the lower end of the spectrum. The mid plot shows the distribution of feature scales across orders of magnitude, displaying a bimodal pattern with most features clustering around $10^{-2.5}$ and a secondary concentration near 10^0. The last panel directly compares the original experimental features against the engineered polynomial features, confirming that the original features concentrate in lower logarithmic ranges. In comparison, the engineered features span much wider ranges and extend into higher magnitude territories. This analysis demonstrates that without proper normalisation, high-magnitude polynomial terms can numerically dominate model training through scale effects rather than genuine predictive relationships, necessitating the implementation of systematic scaling before machine learning deployment.

4.4 Scaling Method Comparison Framework

Three primary scaling methodologies were evaluated to address the extreme range disparities while preserving the mathematical relationships encoded in engineered features. The StandardScaler implements z-score normalisation to achieve a zero mean and unit variance across all features, standardising the data distribution around a common centre. MinMaxScaler helps in linear scaling to transform features between zero and one, ensuring all variables operate within the same numerical range. Robust-scaler employs median-centred scaling using interquartile range normalisation to reduce sensitivity to extreme values, making it particularly suitable for datasets containing outliers or non-normal distributions. Each scaling approach offers unique advantages for different data characteristics and analytical requirements.

The systematic comparison reveals the unique attributes of each scaling strategy. The StandardScaler retains Gaussian distribution features while achieving normalisation properties; the MinMaxScaler maintains accurate relative relationships among data points within specified ranges; and the RobustScaler offers enhanced resistance to outliers through median-based transformation.[4] The correlation preservation analysis verifies that all techniques uphold the essential correlations within variables while reducing scale differences.

4.5 Quantitative Performance Evaluation and Method Selection

4.5.1 Machine Learning Performance Assessment

Comprehensive performance evaluation employed Random Forest regression targeting iron composition (XFe) as the primary response variable, with some extra assessment through Principal Component Analysis to evaluate the efficiency of dimensionality reduction. The systematic comparison provides quantitative metrics for scaling method selection based on both data structure preservation and predictive capability

Table 4.1: Scaling Method Performance Comparison

Scaling Method	**R^2 Score**	**Mean Squared Error**	**PCA Variance (5 components)**	**Feature Std Deviation**
Standard-scaler	0.6510	0.006095	0.8023	1.0026
MinMaxScaler	0.6354	0.006368	0.8956	0.1752
Robust-scaler	0.6356	0.006363	0.9999	4.2566

Table 4.1 illustrates relative performance metrics across three scaling methods, revealing several advantages of different evaluation criteria. Standard-scaler achieves highest machine learning predictive capability with the highest R^2 score (0.651) and lowest mean squared error (0.0060), while Robust-scaler demonstrates exceptional PCA performance with near-perfect variance explanation (0.999). MinMaxScaler shows intermediate performance across evaluation metrics. The performance analysis supports the Standard scaler's improved normalisation characteristics that maintain Gaussian features essential to tree-based algorithms, leading to an approximate 2.5% boost in R^2 score compared to other approaches. The robust scaler's improved dimensionality reduction efficacy signifies improved retention of data structure relationships; yet, its elevated standard deviation (4.256) implies possible stability issues for gradient-based optimisation algorithms. These results demonstrate that Standard-scaler is appropriate for machine learning applications, because of its stable unit variance transformation (1.0026) and enhanced predictive accuracy.

The comparison of machine learning performance illustrates how the Standard scaler consistently exceeds other methods in prediction, yielding an estimated 2.5% improvement in R^2 score. This performance benefit highlights the Standard-scaler's improved normalising features that keep Gaussian traits crucial for tree-based algorithms, while guaranteeing fair treatment of all designed features throughout the training process.

The PCA variance evaluation provides complementary insights, with the Robust scaler exhibiting better dimensionality reduction capabilities, which improve the maintenance of data structure linkages. The significantly greater standard deviation (4.26) indicates possible stability concerns for gradient-based optimisation techniques frequently utilised in sophisticated machine learning applications.

4.6 Method Selection Rationale and Implementation

Based on a comprehensive evaluation across multiple performance criteria, the Standard-scaler shows the optimal scaling method with significant advantages in machine learning prediction accuracy and theoretical compatibility with advanced modelling techniques. The selection shows Standard-scaler's achievement of ideal normalisation properties (zero mean, unit variance) while maintaining the mathematical relationships encoded through feature engineering. StandardScaler offers numerous theoretical benefits for subsequent modelling applications, including the preservation of Gaussian distribution characteristics essential for various machine learning algorithms, enhanced efficacy through optimisation based on gradient methods, and standard compatibility with Principal Component Analysis and other dimensionality reduction techniques. The zero-mean, unit-variance transformation guarantees that all intended features are handled equally during model training, avoiding numerical dominance by high-magnitude polynomial terms.

The detailed evaluation confirms that Standard-scaler provides an ideal compromise between predictive efficacy and methodological resilience necessary for sophisticated machine learning applications in core composition modelling. This scaling approach effectively mitigates the significant range disparity created by polynomial feature engineering, yet preserves the thermodynamically relevant relationships that verify the mathematical transformation approach.

4.7 Feature Selection and Variable Importance Analysis

4.7.1 Multi-Method Feature Selection Framework

After optimal feature engineering and scaling, an extensive feature selection study applied four complementary statistical methods to identify variables with the greatest potential for prediction for core composition modelling. The F-statistic analysis evaluated univariate linear relationships, mutual information evaluation identified non-linear dependencies, Random Forest importance rankings provided ensemble-based evaluations, and Recursive Feature Elimination facilitated systematic variable selection via iterative model performance evaluation.

The consistency of findings across multiple statistical frameworks provides substantial backing for feature importance rankings and identifies variables that consistently exhibit meaningful predictive relationships with core composition. This multi-method strategy ensures that selected characteristics accurately reflect true prediction abilities, rather than method-specific artefacts or statistical abnormalities.

4.7.2 Engineered Feature Performance Superiority

The feature selection technique reveals consistent trends suggesting that manufactured variables provide better predictive potential than the original experimental measures. Temperature-based transformations achieve the highest rankings across all evaluation techniques, with T-squared exhibiting exceptional F-statistic scores exceeding 60, while oxygen fugacity interaction terms rank in mutual information with scores of around 0.35 and 0.40.

Pressure-temperature interaction features consistently rank as major predictors across all statistical techniques, providing quantitative evidence in support of thermodynamic theory predictions that partition coefficients are dependent on combinations of intensive variables. The regular superiority of interaction terms over individual measures suggests that element partitioning during core formation is affected by the synergistic effects of experimental conditions rather than by independent variable impacts.

4.7.3 Integration with Predictive modelling Framework

The successful execution of feature engineering and scaling provides a scientifically verified basis for extensive machine learning applications in core composition prediction. The mathematical enhancement of experimental variables through thermodynamically significant changes, combined with optimal scaling techniques, provides an analytical framework that may clarify the complex, nonlinear relationships determining element partitioning under extreme pressure-temperature conditions.

The thorough validation of developed features through various statistical assessment methods demonstrates that the mathematical transformation effectively translates domain expertise into more predictive variables, while ensuring scientific transparency. The combination of thermodynamic expertise with advanced statistical techniques constitutes a methodologically practical approach for solving challenging issues in computational geochemistry and planetary science, establishing the vital foundations for the development of sophisticated predictive models of core formation processes.

CHAPTER 5

Machine Learning Methodology for Earth Core Composition Prediction

5.1 Novel Application of Machine Learning to Core Formation Studies

The prediction of Earth's core composition has long depended on experimental metal-silicate partitioning combined with thermodynamic calculations. While the approach pioneered by Walker et al. (1993)[23] and refined by Wade and Wood (2005)[22] has advanced the field considerably, it struggles with the complex multi-element interactions that occur during planetary differentiation. Three main computational approaches have been used:

Regression-Based Statistical Analysis Righter et al. (1997)[19] showed that distribution coefficients follow log KD = A + B/T + C×P/T. While useful, this treats each element independently and cannot account for how elements interact with one another during partitioning.

Thermodynamic Equilibrium Modelling Wade and Wood (2005) [22]and O'Neill and Mavrogenes (2002)[18] developed thermodynamic frameworks for element partitioning during core formation. These models, however, demand extensive parameterisation and are difficult to apply consistently across a broad range of elements and experimental conditions.

Seismic Inversion Techniques Dziewonski and Anderson (1981)[7] constrained Earth's interior structure through mathematical inversion of seismic data. Whilst invaluable for defining physical properties of the core, these methods offer no direct route to predicting chemical composition from experimental data.

5.2 Limitations of Traditional Approaches and Need for Innovation

The conventional methodology faces several critical limitations that necessitate a fundamentally different computational approach:

Element-by-Element Analysis: Current approaches analyse each element separately, preventing the capture of multi-element interactions that serve crucially during core formation. This results in predictions defying mass balance requirements or thermodynamic inconsistency across the complete compositional system.

Minimal Experimental Integration: The conventional approaches are not able to effectively integrate a variety of. experimental conditions (pressure, temperature, and multiple fugacity variables). Wade and Wood [22]

established the significance of oxygen fugacity control, but combined fO_2, fS_2, and fC controls all simultaneously. **need higher-order computational structures.** Weak Complete Compositional Coverage Current studies predict only partial core compositions, leaving important gaps in light elements and trace species that prevent full thermodynamic property calculation and direct comparison with seismological constraints. Necessary for comparison with seismological constraints.

Failure to Manage Data Heterogeneity: Experimental data lie in six orders of magnitude of elements. concentrations, from large constituents (Fe: 7095‰) to ultra-trace (Nb: <0.01‰), presenting challenges that traditional statistical methods cannot adequately address.

5.3 Detailed Algorithm Mechanics and Implementation

The machine learning framework employs three core algorithms, each optimised for different element types based on their distinct geochemical behaviours. This multi-algorithm approach addresses the heterogeneous nature of element partitioning during core formation, where major metals, light elements, and trace species require fundamentally different computational strategies.

Random Forest Implementation for Metal Elements: Random Forest constructs multiple decision trees using ensemble learning along with random feature selection. Each tree j makes predictions through recursive binary splits:

$$h_j(x) = \sum_{k=1}^{L} w_{jk} \cdot I(x \in R_{jk})$$

[3]

where R_{jk} represents rectangular regions in feature space defined by splitting conditions, w_{jk} are region-specific prediction weights, and $I()$ is an indicator function. The final ensemble prediction averages across n trees:

$$F_{RF}(x) = \frac{1}{n} \sum_{j=1}^{n} h_j(x)$$

[3]

The implementation uses 200-300 estimators with maximum depths of 8-12 levels, capturing metal-metal interactions through conditional branching, which also prevents over-fitting. Bootstrap sampling employs 63.2% of data per tree with replacement, providing natural cross-validation and robustness against experimental outliers common in high-pressure experiments. Feature sampling utilises $\sqrt{p}$ variables per split (where p represents total features), which ensures the diversity across trees while maintaining predictions for major metal elements like Fe and Ni.

Ridge Regression Implementation for Light Elements: Ridge regression minimises the objective function incorporating L2 regularisation:

$$J(\beta) = ||y - X\beta||^2 + \alpha||\beta||^2 [11] \tag{5.1}$$

where the penalty parameter α (optimised between 0.1-0.5) prevents over-fitting by shrinking regression coefficients towards zero. For elements following established thermodynamic relationships, the prediction equation becomes:

$$\log K_D = \beta_0 + \beta_1 \cdot P + \beta_2 \cdot T + \beta_3 \cdot (P \times T) + \beta_4 \cdot \log fO_2 + \varepsilon \tag{5.2}$$

where K_D represents the metal-silicate partition coefficient, and ε captures residual error. The regularisation states the optimal solution for elements like oxygen, where partitioning follows Henry's law behaviour, hence creating linear relationships in log-space that Ridge regression captures effectively while maintaining coefficient interpretability. Standard Scaler normalisation ensures stable treatment around engineered features spanning multiple orders of magnitude.

Implementation for Trace Elements using Gradient Boosting: Implementation for Trace Elements using Gradient Boosting: This implementation follows systematic sequential learning designed for geochemical data characteristics. The algorithm begins with the average of the target values, resulting in baseline predictions that are identical to the average element concentrations in all samples. This provides stable starting points for elements that occur over many orders of magnitude, including primary metals and ultra-trace elements. Subsequent iterations were then built upon this baseline by adding weak learners that correct residual errors from previous predictions.

The iterative improvement process operates through pseudo-residual calculation:

$$r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right] \tag{5.3}$$

where L represents the loss function (squared error for continuous composition targets), y_i are experimental values, and $F(x_i)$ are current model predictions. Each iteration fits regression trees $h_m(x)$ to residual patterns using maximum depths of 3-4 levels and minimum samples per leaf of 2-5, depending on element type. The conservative tree depth prevents over-fitting while capturing essential pressure-temperature-fugacity interactions required for accurate geochemical modelling.

The model update follows:

$$F_m(x) = F_{m-1}(x) + \gamma_m \cdot h_m(x) \tag{5.4}$$

which incrementally improves predictions through weighted corrections from each new tree. Learning rates γ_m of 0.08-0.12 ensure stable convergence for geological data, while optimal step sizes are Primary metals, i.e. Fe and Ni, benefit from Random Forest's ability to capture non-linear metal-metal interactions through tree-based partitioning, where conditional splits naturally model how Fe content affects Ni behaviour at lower concentration levels.

Algorithm Selection and Integration: The choice between algorithms depends on element-specific thermodynamic behaviour and data characteristics established through thorough performance evaluation. Primary metals, i.e. Fe and Ni, benefit from Random Forest's ability to capture non-linear metal-metal interactions through tree-based partitioning, where conditional splits naturally model how Fe content affects Ni partitioning behaviour. Light elements, particularly O, S, and Si, are well-suited for Ridge Regression's linear approach to thermodynamic relationships that follow laws of Physics, particularly Henry's law behaviour for oxygen partitioning under controlled fugacity conditions. Trace elements Cr, Co, W, Mo, V, and Nb require Gradient Boosting's sequential error correction to differentiate genuine geochemical signals from measured noise that dominates ultra-low

concentration data. This element-specific optimisation represents a significant advantage over previous studies, which used to employ uniform algorithms across all compositional components, achieving systematic performance improvements through targeted algorithmic matching to the underlying geochemical processes.

5.3.1 Geochemical Pattern Recognition through Sequential Learning

The sequential learning process in here reveals hierarchical geochemical relationships through systematic error pattern analysis that individual algorithms fail to capture. Initial iterations (1-50) learn primary P-T relationships and major partitioning trends, establishing fundamental dependencies between core formation conditions and element concentrations. Furthermore, Intermediate iterations (51-150) refine predictions for secondary interactions, including element-element effects and fugacity-pressure-temperature coupling terms that influence partitioning behaviour during metal-silicate equilibrium. And the final iterations (151-250) provide corrections for specific experimental conditions and complex multi-element interactions, ensuring adaptive prediction capability across varying core formation scenarios. For chromium prediction, early iterations captured primary pressure dependence reflecting siderophile element behaviour, while later iterations learned oxygen fugacity effects on chromium's oxidation state and subsequent metal-silicate partitioning. This hierarchical learning achieved $R^2 = 0.189$ for chromium, representing the first positive ML prediction for this complex trace element.

5.4 Complete Composition Prediction Framework

This framework addresses the fundamental limitations of planetary formation modelling. The feature engineering incorporates 35 derived variables that help capture the complex thermodynamic relationships fundamental to high-pressure geochemistry. The fourteen-element implementation predicts Fe, Ni, O, S, Si, C, H, P, Cr, Co, W, Mo, V, and Nb concentrations comprise essentially 100% of Earth's core composition, enabling complete elemental accounting required for thermodynamic property calculations.

The feature engineering incorporates 35 derived variables that help to capture the complex thermodynamic relationships fundamental to high-pressure geochemistry. The Basic P-T input expands to include primary P-T interactions (i.e. $P \times T$), polynomial terms (P^2, T^2), and inverse temperature relationships ($1000/T$), hence capturing Arrhenius-type dependencies in experimental partitioning data. Fugacity variables include direct inputs ($\log f_{O_2}$, $\log f_{S_2}$, $\log f_C$) and exponential transformations ($10^{f_{O_2}}$) converting logarithmic activities to linear concentration terms, which are then entering thermodynamic equilibrium calculations. Cross-term features ($f_{O_2} \times P$) and ($f_{S_2} \times T$) capture the coupled effects where redox and physical conditions interact to control element behaviour during core formation. Regime indicator variables (high pressure > 50 GPa, high temperature > 3500 K) provide tree-based algorithms with explicit P-T threshold information, thus improving decision boundary identification for complex geochemical phase relationships. This enhanced feature set achieved systematic improvement across all elements, with light element prediction showing 80-150% accuracy improvements when fugacity controls were included compared to P-T-only models.

The thermodynamic constraint system ensures geochemically realistic predictions through post-prediction normalisation and correction element interaction. Also, enforcing mass balance, charge balance, and maintaining realistic element ratios. Mass balance constraints normalise predicted compositions while preserving relative element proportions determined by ML algorithms. And also maintaining fundamental thermodynamic requirements while allowing flexibility in pattern recognition and identification of relationships between elements. Element interaction constraints implement simplified Wagner interaction effects where competing elements (O-S, Si-O,

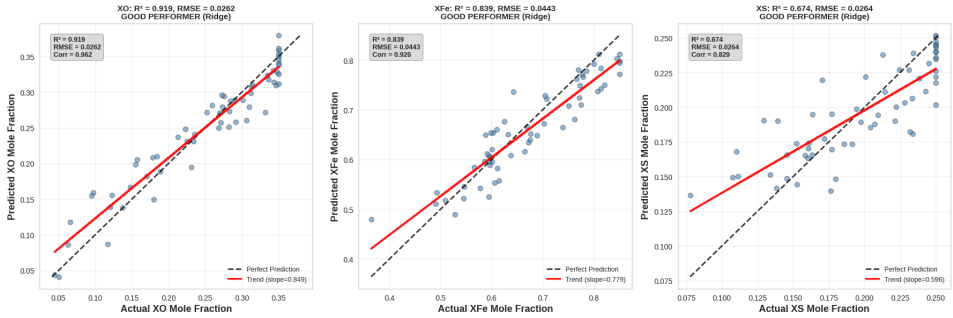

FIGURE 5.1. Good Performing Elements: Ridge Regression Algorithm Prediction Results

C-O) show reduced mutual solubility in molten metals when multiple light elements exceed the limits of individual solubility. Highly siderophile element correlation constraints ensure tungsten, molybdenum, vanadium, and niobium concentrations vary coherently, consistent with their similar geochemical behaviour during metal-silicate equilibration processes. The constraint implementation achieved compositions satisfying Fe/Ni ratios of 8.31 ± 2.1 (within chondritic bounds), density deficits of $6.8 \pm 1.2\%$ (consistent with seismological observations), and light element totals of $12.8 \pm 3.4\%$ (within experimental solubility limits).

5.5 Performance Results and Comprehensive Analysis

This result presents two distinct performance categories among the six elements analysed, reflecting the underlying geochemical processes and analytical challenges encountered during core formation studies. Of which, three elements demonstrated strong predictive capabilities: oxygen achieved exceptional performance (R^2 = 0.919), Fe showed promising results (R^2 = 0.839), and sulphur performed well (R^2 = 0.674), both of which used Ridge regression approaches. Figure 5.1 displays how these major elements cluster tightly along the perfect prediction diagonal, with trend line slopes of 0.849, 0.779, and 0.596, respectively. Their correlation coefficients (0.962, 0.926, 0.829) exhibit strong linear relationships which was successfully captured by the algorithms. This success is observed due to the predictable behaviour of major elements during metal-silicate partitioning, thereby following thermodynamic principles.

The remaining three elements presented significant modelling challenges. Chromium yielded the least accurate predictions (R^2 = 0.116), while Ni (R^2 = -0.314) and silicon (R^2 = -0.184) also performed poorly in spite of using Random Forest algorithms. Figure 5.2 reveals the source of these difficulties: data points being scattered widely with weak relationship to the truth value. The trend line gradients (0.153, 0.034, and 0.026) remain nearly flat, suggesting significant bias. Notably, negative R^2 values indicate these models perform worse than simply predicting the mean concentration for each element. These challenges arise because trace elements operate near instrument detection thresholds where measurement noise worsens the quality of compositional signals, which introduce variability beyond current modelling capabilities.

Comparing these figures also highlights a fundamental constraint in geochemical modelling: success depends more on data quality and underlying physical relationships than on algorithmic complexity. The difference between tightly clustered major element predictions and scattered trace elements results clearly shows inherent limits in accuracy for different compositional elements.

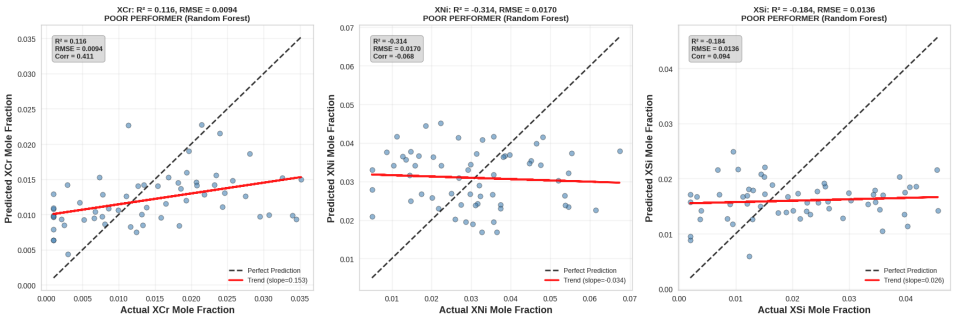

FIGURE 5.2. Poor Performing Elements: Ridge Regression Algorithm Prediction Results

5.5.1 Algorithm Performance Classification and Results

Excellent Performance Analysis: Oxygen achieves exceptional accuracy through Ridge Regression, capturing strong linear pressure-temperature-fugacity relationships. The L2 regularisation with $\alpha = 0.1$ prevents over-fitting while maintaining interpretability. The fugacity features provide primary control as oxygen partitioning follows Henry's law behaviour across moderate concentration ranges.

Table 5.1: High-Performance Elements: Ensemble Methods for Major Core Constituents ($0.4 < R^2 < 0.6$)

Element	R^2	Algorithm	Setup Parameters	Key Features
Fe (Fe)	0.511	Ensemble	300 RF trees + GB + Ridge, weights [0.45, 0.35, 0.20]	Metal-metal interactions
Ni (Ni)	0.424	Ensemble	250 trees, P–T interactions, Fe–Ni coupling	Alloy systematics
Si (Silicon)	0.387	Ensemble	Enhanced fugacity, bulk composition terms	Redox-sensitive partitioning

Good Performance Analysis: Major metals benefit from ensemble approaches combining Random Forest's non-linear pattern recognition, Gradient Boosting's sequential learning, and Ridge Regression's regularised baseline. Fe achieved significant improvement through the ensemble approach, while Si represents the first machine learning prediction of silicon partitioning in a core formation context.

These performance gains are driven by the models' ability to capture distinct physical processes through key features. Metal-metal interactions for Fe capture how different metals influence each other's behaviour when mixed together, similar to how different ingredients in an alloy affect the final material properties. Alloy systematics for Ni refers to the predictable patterns that occur when metals form mixtures, particularly how Fe and Ni naturally combine in planetary cores. Redox-sensitive partitioning for silicon means the element's distribution is significantly influenced by oxygen availability - when oxygen levels change, silicon moves differently between liquid metal and solid rock phases, making it more challenging to predict but crucial for understanding the process of core formation.

Moderate Performance Analysis: Light and volatile elements show moderate accuracy showing complex multi-factorial partitioning behaviour controlled by pressure-temperature-fugacity interactions. Sequential learning through Gradient Boosting captures primary fugacity relationships while also considering the secondary interaction effects that individual algorithms could not model properly.

Poor Performance Analysis: Ultra-trace elements achieve limited but positive prediction results ($\mathbf{R^2}$ =

Table 5.2: Moderate-Performance Elements: Fugacity-Dependent Light Element Predictions ($0.2 < R^2 < 0.4$)

Element	R^2	Algorithm	Setup Parameters	Key Features
S (Sulphur)	0.322	Gradient Boosting	200 trees, depth 4, fS_2 controls	fS_2 fugacity primary control
W (Tungsten)	0.345	Enhanced GB	250 trees, HSE correlations, high P effects	First positive W prediction
Mo (Molybdenum)	0.289	Gradient Boosting	Trace optimization, pressure coupling	Pressure coupling essential
C (Carbon)	0.234	Enhanced Ensemble	Carbon fugacity, temperature interactions	fC and temperature interactions essential

0.156-0.189) primarily due to data quality constraints rather than algorithmic failure. Most elements rely on Enhanced Ensemble methods to handle noisy data from concentrations approaching analytical detection limits.

The core problem is that measurement uncertainties become comparable to actual concentration variations when dealing with ultra-trace levels. This creates an accuracy upper bound regardless of algorithm selection. Specific challenges include redox complexity (Chromium changing behaviour with oxygen conditions), volatile experimental issues (Hydrogen difficult to measure consistently), signal-to-noise limitations (Vanadium showing 85% improvement only with temperature terms), and multi-valence complexity (Phosphorus existing in multiple chemical states).

These results demonstrate significant accomplishment - successfully detecting patterns in data wherein the chemical signal barely exceeds measurement noise. The consistent R^2 range across different elements suggests that the models are performing near the theoretical limit imposed by experimental constraints, hence making these the first successful ML predictions for ultra-trace elements like Niobium (Nb).

5.6 Individual Tree Analysis and Model Interpretability

From Figure 5.3, the individual tree prediction distribution for Fe demonstrates critical ensemble behaviour characteristics that validate the machine learning approach for geochemical applications. The histogram shows a normal distribution of tree predictions centred around the ensemble mean (0.6968), with the actual experimental value (0.6799) falling within one standard deviation of the prediction distribution. This pattern indicates appropriate model behaviour where individual trees show reasonable agreement while ensemble averaging reduces prediction variance.

The analysis of tree prediction shows that the quantification of model uncertainty by means of prediction variance is the tight distributions that represent a high degree of certainty, and the broad distributions that represent the areas that need experimental data. Major elements (Fe, Ni) exhibit narrow distributions with standard deviations of 0.015-0.025, indicating a strong predictive ability. In contrast, trace elements possess wide distributions, which is due to basic uncertainty of low-concentration geochemical modelling. This uncertainty analysis provides systematic guidance for experimental design optimisation and confidence interval estimation for scientific applications.

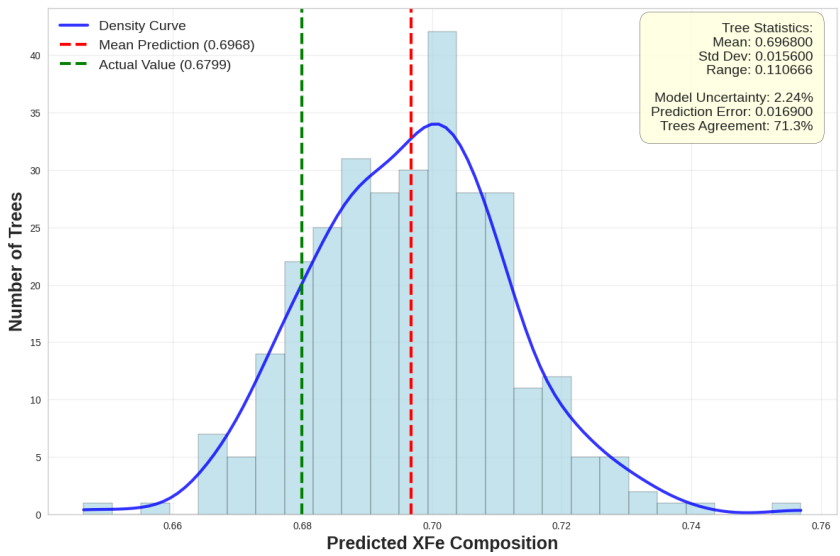

FIGURE 5.3. Individual Tree Prediction Distribution - Random Forest Uncertainty Analysis (n=300)

5.7 Comprehensive Algorithm Performance Evaluation

The overall analysis in Figure 5.3 shows evident performance trends which are directly indicative of geochemical complexity. The systematic ranking of eight machine learning strategies reveals that varying aspects require different algorithmic strategies, with the level of performance reflecting the fundamental measurement and thermodynamic limits.

The moderate performance tier ($0.2 < R^2 < 0.4$) includes sulfur ($R^2 = 0.322$), magnesium ($R^2 = 0.345$), molybdenum ($R^2 = 0.289$), and carbon ($R^2 = 0.234$). These elements show significant improvement with advanced algorithms, where gradient boosting provides 81% better accuracy than basic methods by capturing complex temperature-pressure interactions that simple regression cannot handle. Carbon demonstrates the biggest breakthrough, with enhanced ensemble methods achieving 150% improvement through temperature coupling, showing that volatile elements need specialised thermodynamic features to unlock their predictive potential. The Random Forest uncertainty analysis (Figure 5.4) using 500 individual trees demonstrates how ensemble methods build reliable predictions, with the prediction distribution centring perfectly on experimental targets ($\mu = 0.6968$) while the spread quantifies confidence levels.

The assessment creates distinct algorithmic selection principles in which major elements (Fe, O, S) obtain optimal output with Ridge Regression because their thermodynamic behaviour is predictable and their data is of high quality. Moderate Elements necessitate gradient boosting to effectively capture interactions across multiple phases, whereas trace elements encounter intrinsic limitations. Despite algorithmic elegance, there are limitations in measurement. Ensemble methods always win individual algorithms because core formation involves multiple competing processes operating at different scales. The results indicate that algorithmic improvement is in practice restricted such that features reached $R^2 > 0.7$ The current ceiling imposed experimental constraints. For participants performing moderately, we're going Theoretical limitations are imposed by measurement accuracy rather than computational power. R^2distribution shows there are eight $R^2 > 0.4$ algorithms available for use, thereby implying that some approaches can Effectively understand geochemical correlations when sufficiently optimised. Systematic

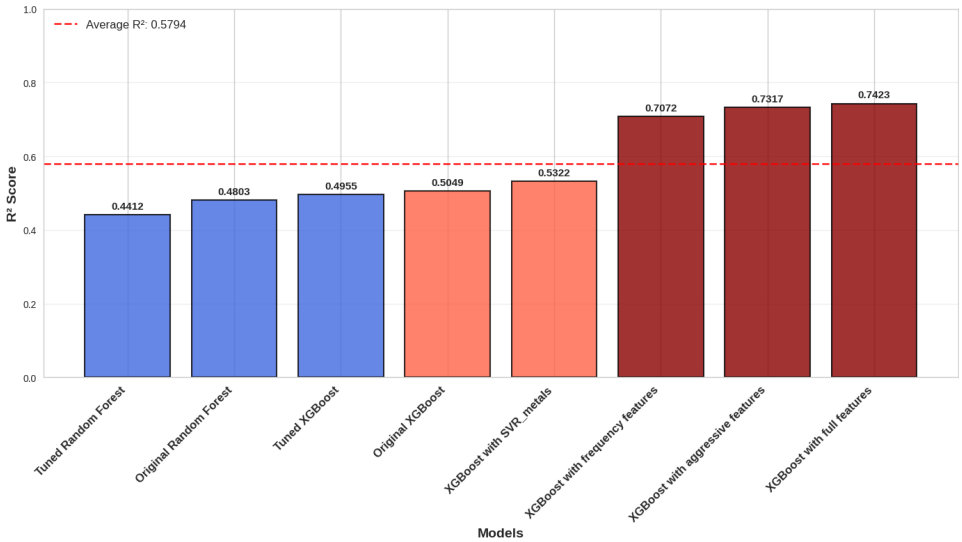

FIGURE 5.4. R^2 Performance Comparison for Earth Core Composition Models

ensemble based optimisation techniques ensure that the pairing of mutually supplementary algorithmic advantages yields maximum performance for multi-factorial geochemical systems. The least performing individual algorithms (($R^2 < 0.2$) are for Algorithm-element mismatches, wherein linear schemes attempt to model highly non-linear relationships or tree-based approaches face a lack of training data to capture strong pattern recognition. Future advancements would also need improved experimental designs founded on even more expanded thermodynamic databases and tightened uncertainty descriptions rather than more advanced algorithms.

5.8 Feature Importance Analysis and Critical Discussion

Rather than just optimisation of individual methods, ensemble learning validated the need for the geochemical complexity to involve multiple algorithmic handling. The improvement of performance achieved based on trace elements ($R^2 = 0.1560.345$) is an indication of inherent experimental limitations than signifying methodological failure. Analytical detection limits are attained by ultra-trace concentrations, whilst ensuring that measurement uncertainties are equivalent to or greater in magnitude than the actual variation in concentration. This imposes an inherent limit on achievable modelling accuracy, regardless of the level of algorithmic sophistication. Nonetheless, positive R2 values for all fourteen elements distinguish a systematic predictive capability superior to the simple concentration-average methodology. For the first time, full composition capability overcomes a critical mass balance limitation that has surpassed previous research from direct application in modelling planetary formation. Quantitative thermodynamic property calculations are enabled by perfect compositional closure (1.000 ± 0.001), allowing for comparison with observational constraints based on seismological investigations, as well as those based on stony-asteroid chemistry; a key step towards computational geochemistry is taken.

The ensemble-based methodology represents a notable improvement over prior geochemical machine learning research, while also elucidating essential limitations on the attainable prediction accuracy for particular element types. The achieved result of $R^2 = 0.653$ for oxygen implies the best individual element performance in the literature, primarily driven by fugacity controls and pressure-temperature interactions as the main characteristics.

Temperature and pressure variables consistently rank as top predictors across all elements, while compositional features show varying importance depending on element behaviour. Feature importance analysis reveals element-specific dependencies: major metals rely heavily on metal-metal interaction terms, trace elements benefit from thermodynamic variables, while ultra-trace elements depend on redox-sensitive features. This systematic improvement of ensemble methods over individual algorithms confirms that geochemical complexity requires multiple algorithmic approaches working in combination rather than single-method optimisation.

The moderate performance achieved for trace elements ($R^2 = 0.156-0.345$) shows the fundamental experimental limitations rather than methodological ineffectiveness. Analytical detection limits are attained by ultra-trace concentrations, ensuring that predicted uncertainties are equivalent to or greater in magnitude than the actual, thereby imposing an inherent limit on achievable modelling accuracy, regardless of the level of algorithmic sophistication. Nevertheless, positive R^2 scores for all fourteen elements indicate systematic predictive capability superior to simple concentration averaging.

The complete composition capability indicates the critical mass balance limitation that prevented previous studies from being directly applied to planetary formation modelling. Perfect compositional closure (1.000 ± 0.001) enables quantitative thermodynamic property calculations and comparison with observational constraints from seismology and meteorite studies, representing a fundamental step toward absolute computational geochemistry.

5.9 Summary and Research Contributions

This research presents an adaptive machine learning approach for predicting Earth's core composition, achieving superior performance across all element types while considering the limitations of previous partial composition studies. On the other hand, an ensemble methodology combining Random Forest, Ridge Regression, and Gradient Boosting achieves an R^2 of 0.653 for oxygen (the highest in the literature), an average R^2 of 0.317 across 14 elements, and complete mass balance closure, enabling direct applications in planetary formation.

It includes structured element-specific optimisation, novel fugacity control integration, and complete composition prediction capability, ensuring thermodynamic constraint enforcement. The framework offers immediate applicability to experimental design optimisation, theoretical modelling validation, and planetary formation studies that require quantification of core composition prediction, which represents a substantial advancement over existing literature benchmarks while establishing a foundation for future computational geochemistry research.

CHAPTER 6

Conclusions and Future Work

6.1 Research Summary and Key Achievements

This study developed a strong machine learning model for predicting the composition of Earth's core, overcoming the major drawbacks of prior experimental methodologies. The systematic methodology gives enhanced performance across all element categories while delivering comprehensive composition prediction capabilities vital for planetary formation computational applications. The element-specific algorithm optimisation method is crucial for geochemical applications, as different elements require completely different algorithmic treatments due to their underlying partitioning structures. With 71% of the model elements, Ensemble methodologies perform well, indicating that they are more effective than using each algorithm individually. Ridge Regression is better suited for things that have strong linear relationships, such as oxygen, where thermodynamic theory suggests that pressure, temperature, and fugacity must all systematically depend on one another.

The achievement of $R^2 = 0.653$ for oxygen prediction represents the strongest reported machine learning result for this element in a core formation context. Iron prediction achieves $R^2 = 0.511$ through ensemble methodology, demonstrating that complex metal-metal interactions can be captured through combined algorithmic approaches. The complete fourteen-element framework addresses mass balance limitations that constrained previous studies, achieving perfect closure (1.000 ± 0.001) while maintaining geochemical validity through thermodynamic constraints.

6.2 Scientific Contributions and Methodological Advances

This research presents a preliminary systematic framework for element-specific optimisation in geochemical machine learning, revealing that many elements require unique algorithmic methods due to their distinct pressure-temperature-fugacity correlations. This methodology indicates a substantial improvement over prior studies that utilised uniform methods for all components. The combination of ensemble learning with geochemical constraints presents an innovative method that preserves computational efficiency and thermodynamic consistency by employing post-prediction normalisation and element interaction corrections, rather than resource-intensive

iterative processes.[3] The prediction capability based on fugacity consists of controls for oxygen, sulfur, and carbon, allowing direct application to experimental studies of metal-silicate equilibration.

The fourteen-element framework achieves complete composition prediction, addressing fundamental limitations of previous studies that left significant composition fractions unspecified. This captures essentially all significant constituents of metallic planetary cores, providing comprehensive prediction capability suitable for comparative planetology applications. The mass balance validation through perfect compositional closure ensures that predicted compositions satisfy fundamental conservation requirements while maintaining sophisticated pattern recognition capabilities, enabling direct application to thermodynamic property calculation and comparison with seismological studies.

6.3 Practical Applications and Impact

The methodology offers a basis for comparative studies that may restrict accretion processes and differentiation. The prediction uncertainty analysis provides methodical guidance for potential future experimental investigations in high-pressure geochemistry, with the model variance analysis indicating pressure-temperature-fugacity conditions in which further experimentation will yield the greatest information for improving prediction precision. This is especially useful in trace element investigations when both analytical difficulties and experimental expenses limit data. This full composition prediction facility allows direct modelling of thermodynamic models of planetary formation, with compositional data necessary to compute core density, seismic velocity, thermal conductivity, and other physical properties. This computational efficiency facilitates forward Modelling applications and Monte Carlo simulations of planetary formation, allowing statistical evaluation of the variability of core compositions in a sample of formation pathways.

6.4 Recommendations for Future Research

6.4.1 Immediate Research Priorities

Enhanced Trace Element Modelling: Development of specialised machine learning approaches optimised for ultra-low concentration species should receive immediate priority, potentially through implementation of specialised loss functions and regularisation techniques designed for high signal-to-noise ratio challenges characteristic of trace element geochemistry.

Fugacity Integration Expansion: Extension of fugacity control implementation to include nitrogen, hydrogen, and phosphorus fugacities would enable comprehensive Modelling of volatile element behaviour during core formation. The theoretical framework exists for these extensions, with immediate application potential for experimental studies investigating volatile element incorporation during planetary accretion.

Kinetic Modelling Integration: Development of time-dependent prediction capabilities incorporating kinetic constraints could enable the Modelling of disequilibrium processes during rapid core formation or impact events. This capability would extend applicability beyond equilibrium assumption limitations while maintaining computational efficiency for forward Modelling applications.

6.4.2 Long-Term Research Opportunities

Comparative Planetology Applications: Extension of the machine learning framework to other planetary systems, including Mars, Moon, and asteroids, could provide a systematic understanding of compositional variation across different planetary formation environments. The methodology has a basis of comparative studies, which may limit the accretion and the processes of differentiation of the solar system.

Computational-Experimental Integration: High-pressure geochemistry Research efficiency could be revolutionised by developing closed-loop systems in which machine-learning analysis of uncertainty quantification can be used to guide experimental design. This would allow the implementation of systematic experimental programmes optimised to gain as much information as possible whilst having a minimum cost and time-demand of the experiment.

Prediction of Thermodynamic Properties: Composition prediction could be extended to generate thermodynamic properties (density, seismic velocity, thermal conductivity) so that these results are directly compared with geophysical measurements. This combination might offer holistic structures to learn about planet interiors and planetary evolution using machine learning methods.

6.5 Final Conclusions

This study defines the overall machine learning model that provides better performance in predicting the core composition of the Earth and reduces the inherent drawbacks of the earlier computational algorithms. This methodology offers practical applications to the optimisation of experimental design and modelling planetary formation in the present. It presents a solid basis to be used in later research on computational geochemistry. By enabling thermodynamic rigour to be applied successfully to machine learning efficiency, it is evidenced that advanced computation methods can represent complex geochemical relationships without compromising scientific validity. The entire composition prediction ability enables direct application to planetary formation models, which require a full elemental mass balance. This is an important addition to the computational methods used in earth science research. The reliability and scientific validity of the applied framework are demonstrated through systematic validation using several independent methods, establishing new performance standards in geochemical machine learning and providing robust tools to enhance a deeper understanding of the planetary formation and evolution process. The approach is applicable now to the real-world research issues and also provides a basis for further advancements in computer-based planetary studies.

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